

**Mécanismes Moléculaires de la Signalisation Cellulaire (MMSC)
Master de Biologie Intégrative, M1S1
TD / Exercices**

Small G proteins

Dr Pierre JOANNE (pierre.joanne@sorbonne-universite.fr)

Pr Isabelle LIMON (isabelle.limon@sorbonne-universite.fr)

Dr Luce PERIE ((luce.perie@sorbonne-universite.fr)

Prerequisites: Wooclap

Group exercise

The method of analysis

The classical way to analyze figures

1. Question → 2. Material → 3. Methods → 4. Results → 5. Conclusion → 6. Perspective & discussion

The table for the analysis of figures

Item	Example
What models are used?	A single model: sea urchin eggs
What are the experimental conditions?	Condition 1: Injection of different inositol phosphates into the eggs Condition 2: Each inositol phosphate is injected at different doses Condition 3: Eggs are cultured, after injection, either in normal medium or in calcium medium
What is being measured?	A single measurement: the % of eggs showing a complete exocytosis of cortical granules
(Optional) How are the measurements made? or How is the experiment carried out?	Manual counting under a microscope of the number of eggs showing complete exocytosis of cortical granules, compared to the total number of eggs (including those showing no response)
What is observed?	Observation 1: IP3 is the inositol phosphate that triggers 100% exocytosis at the lowest dose Observation 2: Dose-dependent effect of inositol phosphate injection → the higher the concentration, the higher the % of eggs showing complete exocytosis of cortical granules Observation 3: No difference between the two culture media
What interpretation can be made from these observations?	Injection of inositol phosphate mimics the first stage of fertilization, and IP3 is the most effective phosphate to induce cortical granule exocytosis. It therefore appears to be a key player in this physiological function, which seems to be independent of extracellular calcium.
So, what question does this figure answer?	Several questions: Is inositol phosphate a player in fertilization? If so, which one is most effective? Does its function require the presence of extracellular calcium?
What new question does this raise?	A bit of imagination is needed. For example, one might wonder whether IP3 injection could go further and trigger the early stages of embryonic development, to verify whether inositol phosphate is also involved in that physiological phenomenon.

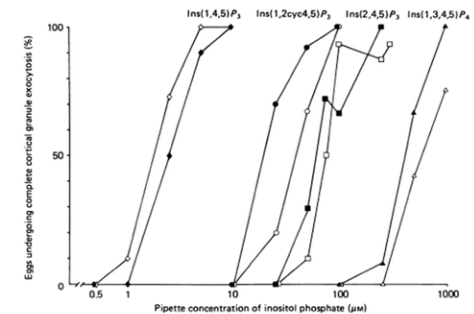


Figure 1: Inositol phosphate-induced cortical granule exocytosis.

Eggs (sea urchin) were microinjected with a constant volume of inositol phosphate (1.5 μ l) and the pipette concentration was varied. The eggs were scored for cortical granule exocytosis by direct observation of the cortical granules by using differential interference contrast microscopy. Closed symbols refer to injections performed in normal ASW. Open symbols refer to injections performed in calcium-free ASW. Each point on the graph represents the injection of between 10 and 15 eggs.

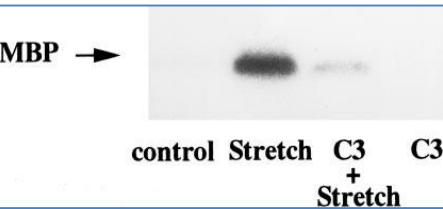


Figure 1 : Cardiomyocytes were stretched by 20% for 9 minutes (Stretch conditions). Cardiomyocytes were treated for 36 hours with the exoenzyme C3 (blocker of the activity of Rho family proteins. NB: C3 does not inhibit the basal activity of ERK). From the cell lysate of these cardiomyocytes, ERK is immunoprecipitated and its activity is measured in the presence of MBP (peptide substrate of ERK1 and ERK2) and $[\gamma\text{-}^{32}\text{P}]\text{ATP}$. After electrophoresis the gel is revealed by autoradiography.

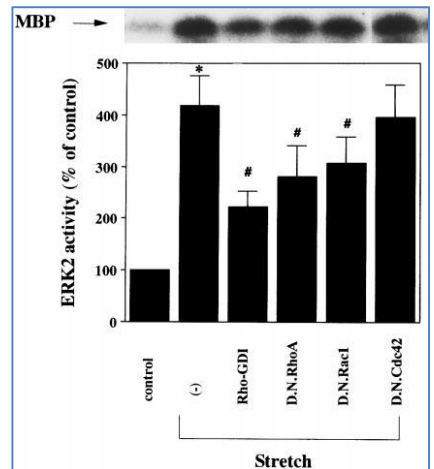


Figure 2 : Cardiomyocytes were co-transfected with HA-ERK2 and Rho-GDI, or D.N.RhoA, or D.N.Rac1, or D.N.Cdc42. These cardiomyocytes were stretched and the resulting lysate was immunoprecipitated with an anti-HA antibody. ERK activity was then measured as previously and quantified. Data are reported relative to control. * $P < 0.05$ versus control. # $P < 0.05$ versus stretch (-).

D.N. = Dominant Negative

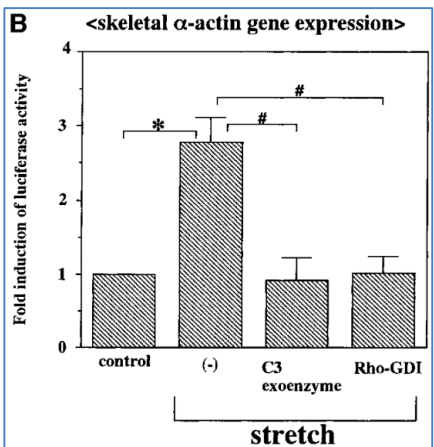


Figure 3 : A reporter gene (Luciferase) of skeletal actin expression was transfected into cardiomyocytes and measured on a luminometer. Cardiomyocytes were stretched and treated with exoenzyme C3 or transfected with Rho-GDI. * $P < 0.05$ versus control. # $P < 0.05$ versus stretch (-).

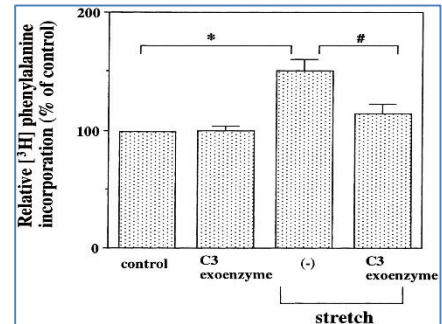


Figure 4 : After pretreatment with 10 $\mu\text{g}/\text{mL}$ C3 exoenzyme for 36 hours, cardiac myocytes were stretched 20% for 24 hours and $[\text{H}^3]\text{phenylalanine}$ (1 $\mu\text{Ci}/\text{mL}$) was added 2 hours before harvest. Total radioactivity of $[\text{H}^3]\text{phenylalanine}$ incorporated into proteins was determined by liquid scintillation counting. Data represent the mean percentage of controls (100%) from 3 independent experiments (mean $\pm$ SE). * $P < 0.05$ versus control and # $P < 0.05$ versus stretch.

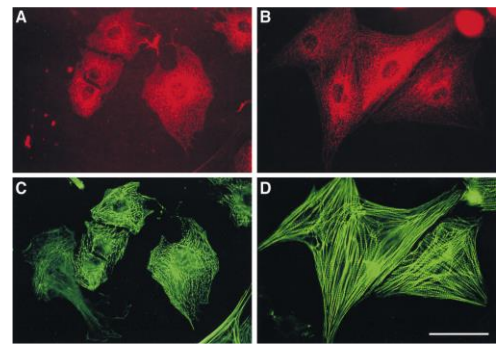


Figure 5 : Effects of Ad.V12rac1 or Ad.N17rac1 expression on sarcomeric organization and surface area of cultured cardiomyocytes. A-D: Myocyte morphology 48 h after infection with (A and C) a control adenovirus (Ad.dl312) or (B and D) Ad.V12rac1 (adenovirus inducing the expression of a constitutively active Rac1 mutant). Cells were analyzed by immunofluorescence with a myosin-specific antibody (sarcomere marker, red). Actin filaments were visualized using fluorescein-conjugated phalloidine (green). E: Quantification of myocyte cell surface area after infection with a MOI (multiplicity of infection) of 25 or 100 with either Ad.V12rac1 or Ad.N17rac1 (adenovirus inducing the expression of a dominant-negative Rac1 mutant). Cells were measured under basal conditions or following a hypertrophic treatment (Phenylephrine). * $P < 0.05$ versus untreated control or treated control.

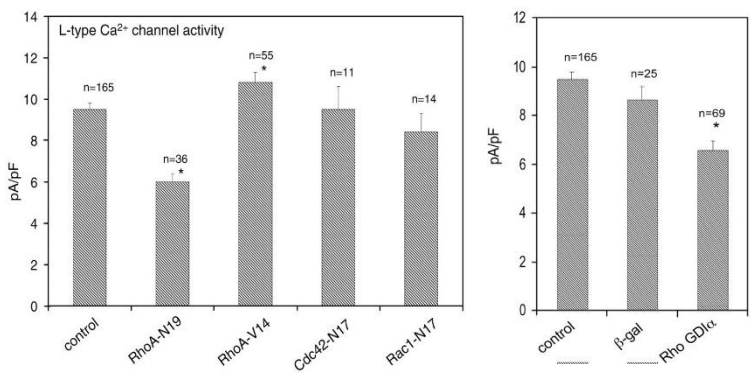
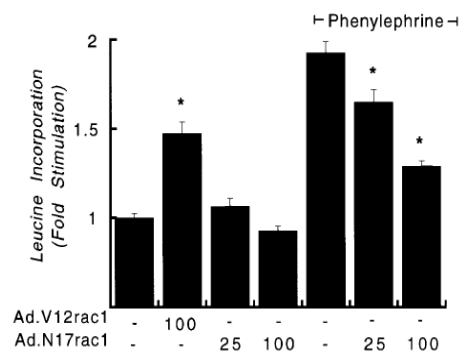


Figure 6 : Effects of Rho family protein modulation on ICa in adult mouse ventricular myocytes. The holding potential was -50 mV and the test pulse was +10 mV. A: Consequences of acute expression of dominant-negative mutants of RhoA, Rac-1, and Cdc42 (RhoA-N19, Rac-1-N17, and Cdc42-N17, respectively) and a constitutively active mutant of RhoA (RhoA-V14) in cardiomyocyte cultures through TAT fusion protein delivery. B: Consequences of expression of beta-galactosidase and Rho-GDI α in cardiomyocyte cultures through TAT fusion protein delivery. * $P < 0.01$ compared with untreated (control) myocytes.

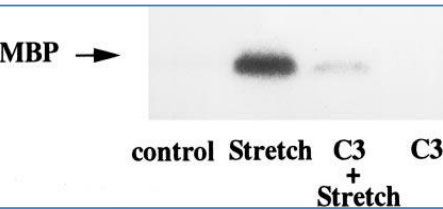


Figure 1 : Des cardiomyocytes ont été étirés de 20 % pendant 9 minutes (conditions Stretch). Des cardiomyocytes ont été traités pendant 36 heures avec l'exoenzyme C3 (bloqueur de l'activité des protéines de la famille Rho. NB : C3 n'inhibe pas l'activité basale de ERK). A partir du lysat cellulaire de ces cardiomyocytes, ERK est immunoprécipité et son activité est mesurée en présence de MBP (peptide substrat de ERK1 et ERK2) et de $[\gamma\text{-}^{32}\text{P}]\text{ATP}$. Après électrophorèse le gel et révélé par autoradiographie.

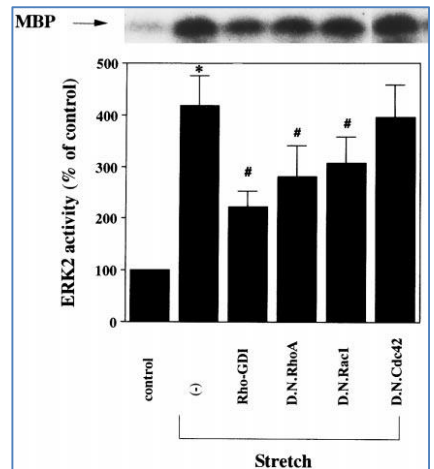


Figure 2 : Des cardiomyocytes sont co-transfectés avec HA-ERK2 et Rho-GDI, ou D.N.RhoA, ou D.N.Rac1, ou D.N.Cdc42. Ces cardiomyocytes sont étirés (Stretch) et le lysat obtenu est immunoprécipité avec un anticorps anti-HA. L'activité ERK est ensuite mesurée comme précédemment et quantifiée. Les données sont rapportées par rapport au contrôle. * $P < 0,05$ versus control. # $P < 0,05$ versus stretch (-).

D.N. = Dominant Négatif

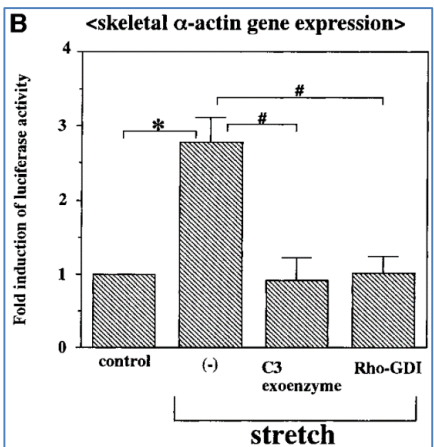


Figure 3 : Un gène rapporteur (Luciférase) de l'expression de l'actine squelettique est transfecté dans des cardiomyocytes et est mesurée sur un luminomètre. Les cardiomyocytes sont étirés (stretch) et traités avec l'exoenzyme C3 ou transfectés avec Rho-GDI. * $P < 0,05$ versus control. # $P < 0,05$ versus stretch (-).

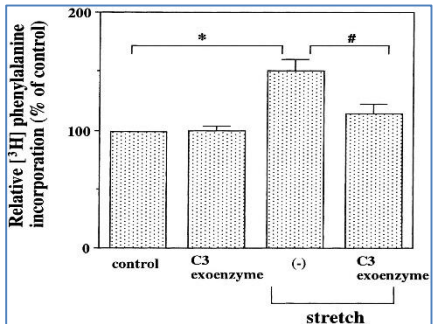


Figure 4 : Après un prétraitement avec 10 $\mu\text{g/mL}$ d'exoenzyme C3 pendant 36 heures, les myocytes cardiaques ont été étirés de 20 % pendant 24 heures et de la $[\text{H}^3]\text{phénylalanine}$ (1 $\mu\text{Ci/mL}$) a été ajoutée 2 heures avant la récolte. La radioactivité totale de la $[\text{H}^3]\text{phénylalanine}$ incorporée dans les protéines a été déterminée par comptage par scintillation liquide. Les données représentent le pourcentage moyen de contrôles (100 %) de 3 expériences indépendantes (moyenne $\pm$ SE). * $P < 0,05$ par rapport au contrôle et # $P < 0,05$ par rapport à l'étirement (stretch).

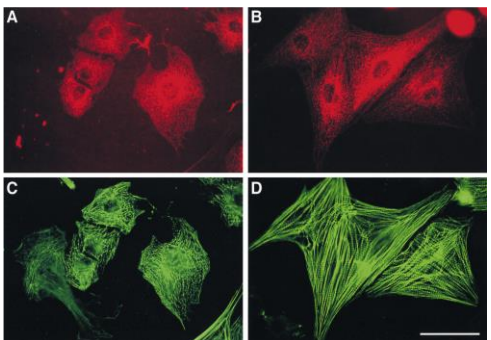


Figure 5 : Effets de l'expression d'Ad.V12rac1 ou Ad.N17rac1 sur l'organisation sarcomérique et la surface des cardiomyocytes en culture. **A-D :** Morphologie des myocytes 48 h après l'infection par (A et C) un adénovirus témoin (Ad.dl312) ou (B et D) l'Ad.V12rac1 (adénovirus induisant l'expression d'un mutant Rac1 constitutivement actif). Les cellules ont été analysées par immunofluorescence avec un anticorps spécifique à la myosine (marqueur des sarcomères, rouge). Les filaments d'actine ont été visualisés en utilisant de la phalloïdine conjuguée à la fluorescéine (vert). **E :** Quantification de la surface cellulaire des myocytes après infection avec une MOI (multiplicity of infection) de 25 ou 100 soit avec Ad.V12rac1 soit avec Ad.N17rac1 (adénovirus induisant l'expression d'un mutant dominant négatif de Rac1). Les cellules ont été mesurées en conditions basales ou suite à un traitement hypertrophiant (Phényléphrine). * $P < 0,05$ versus untreated control or treated control.

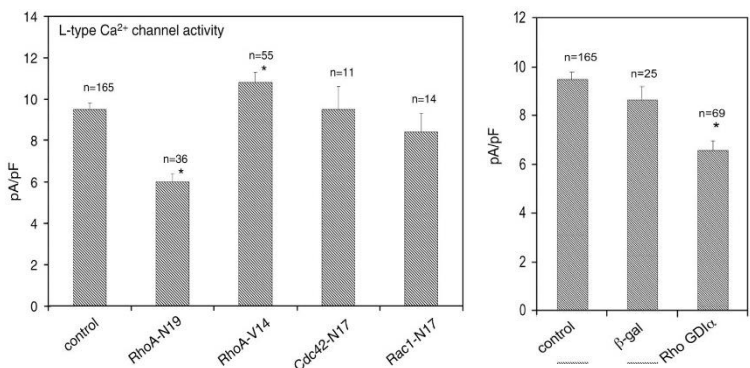
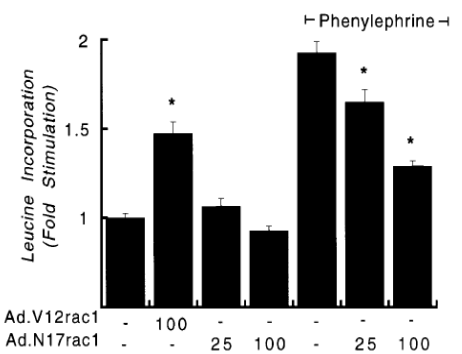
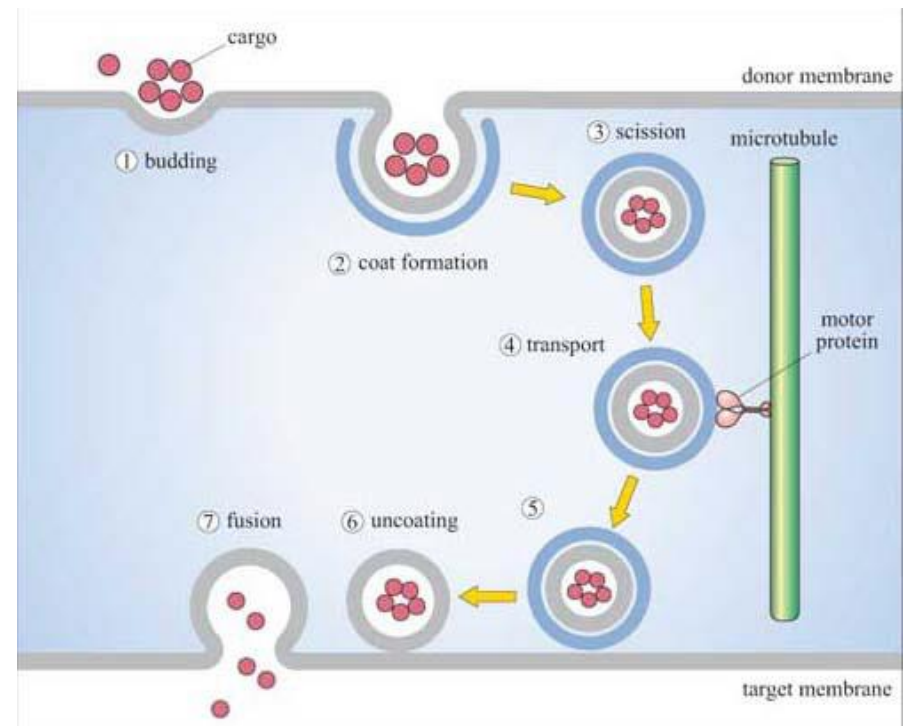
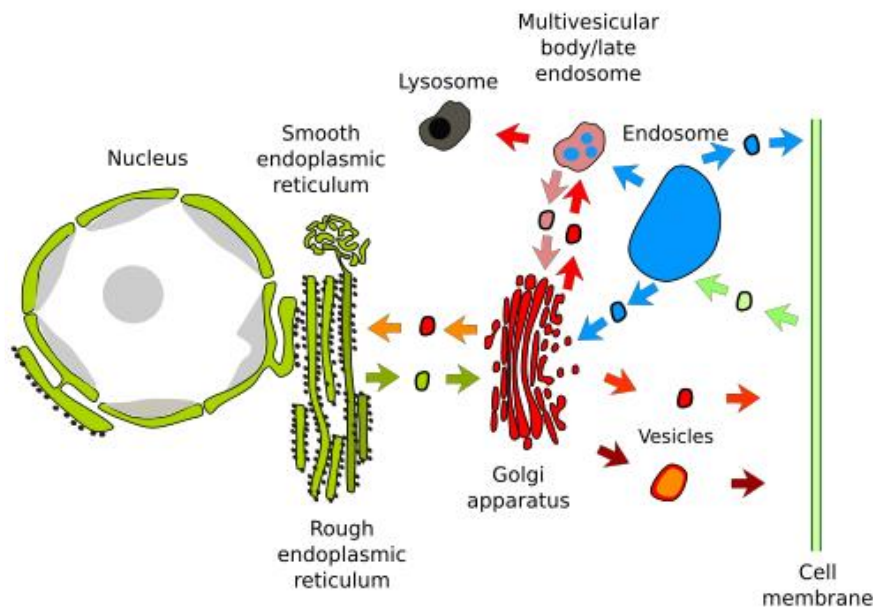


Figure 6 : Effets de la modulation des protéines de la famille Rho sur ICa dans les myocytes ventriculaires de souris adultes. Le potentiel de maintien était de -50 mV et l'impulsion de test était de +10 mV. **A :** Conséquences de l'expression aiguë de mutants dominants négatifs de RhoA, Rac-1 et Cdc42 (RhoA-N19, Rac-1-N17 et Cdc42-N17, respectivement) et d'un mutant constitutivement actif de RhoA (RhoA-V14) dans des cultures cardiomyocytes grâce à la délivrance par protéine de fusion TAT. **B :** Conséquences de l'expression de la bêta-galactosidase et de Rho-GDI α dans des cultures cardiomyocytes grâce à la délivrance par protéine de fusion TAT. * $P < 0,01$ par rapport aux myocytes non traités (control).

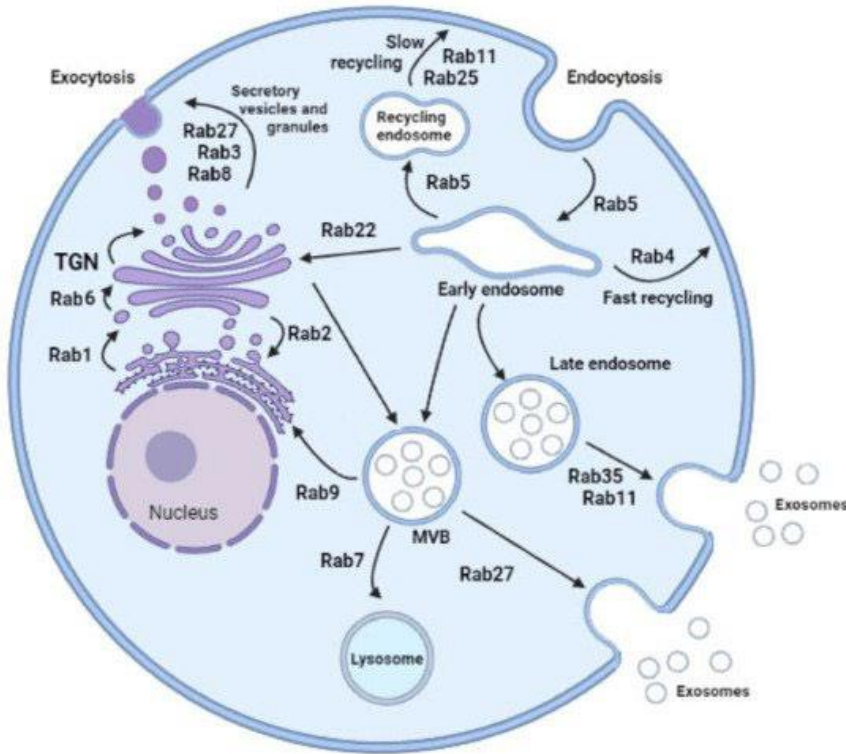
Exercise: An example of small G protein in vesicle trafficking

Vesicle trafficking

In eukaryotes, intracellular vesicle trafficking allows the exchange of molecules between different compartments. The trafficking of these vesicles is finely regulated: it involves a control of their formation in the donor compartment, their transport to the recipient compartment along the cytoskeleton and their recognition by the recipient compartment.



Rab family



Numerous studies have shown that the monomeric G proteins of the Rab family regulate the targeting of these vesicles towards the acceptor compartment. On the other hand, their involvement in the formation of these vesicles is not established.

In their study, S. Miserey-Lenkei et al (S. Miserey-Lenkei, et al., Nat. Cell. Biol. 12 : 645-654, 2010) proposed to elucidate the role of two ubiquitous Rab6A isoforms (Rab6A and Rab6A') in this crucial step.

Their results are presented below.

Experimental data

The authors first assessed the impact of Rab6A and Rab6A' on Rab6-associated transport vesicles using interfering RNA in HeLa cells. The results are shown in Figure 1.

siRNA

Nanoparticle
vector

siRNA

RISC

RISC
siRNA

RISC
siRNA

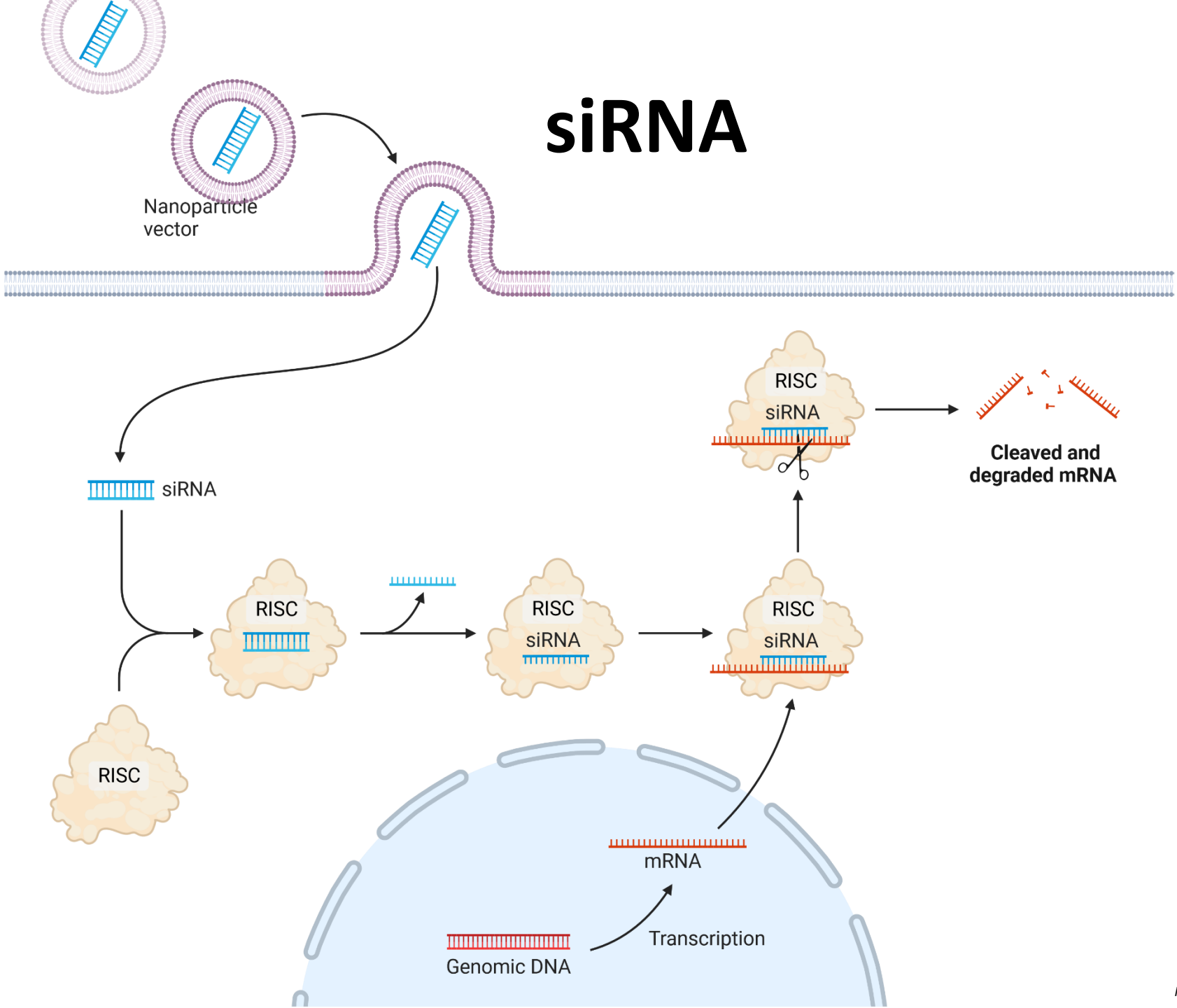
RISC
siRNA

Cleaved and
degraded mRNA

mRNA

Transcription

Genomic DNA



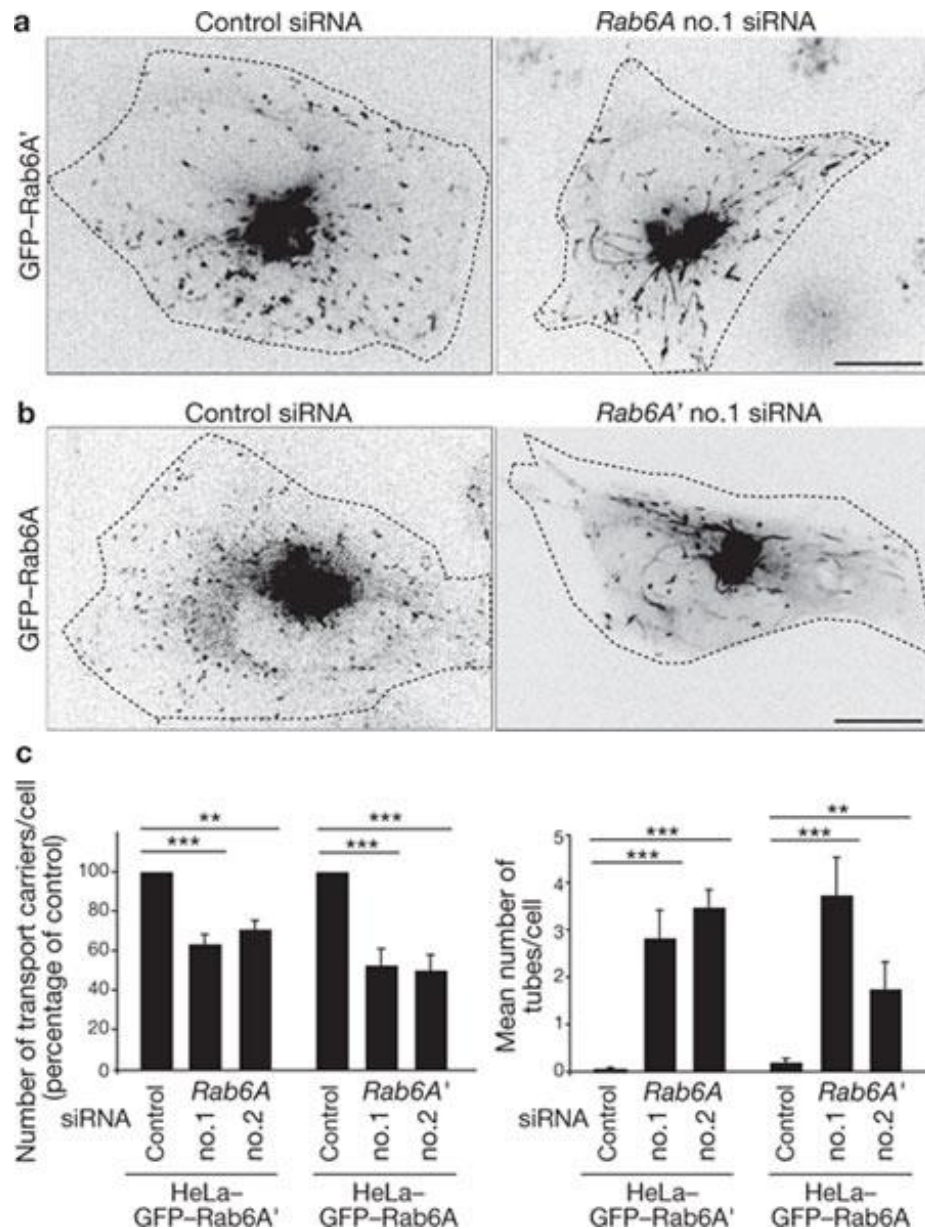


Figure 1. (a) HeLa cells expressing GFP-Rab6A' were treated for 3 days with control or Rab6A-specific siRNAs. Cells were visualized by fluorescent microscopy (b) HeLa cells expressing GFP-Rab6A were treated for 3 days with control or Rab6A' specific siRNAs. Scale bars, 10 μ m. (c) The number of transport carriers (left panel) and unfissioned tubules connected to the Golgi (right panel) are shown for each condition (mean $\pm$ s.e.m., n = 21–42 cells). Two independent (no. 1 and no. 2) isoform-specific siRNAs against Rab6A or against Rab6A' were used. **P < 10⁻³ and ***P < 10⁻⁵ (Student's t-test).

Figure 1

Among the many potential partners of Rab6, myosin II is a possible candidate in the initiation step of vesicle formation. The authors therefore investigated the possible relationship between Rab6 proteins and myosin II. The corresponding data are shown in Figure 2.

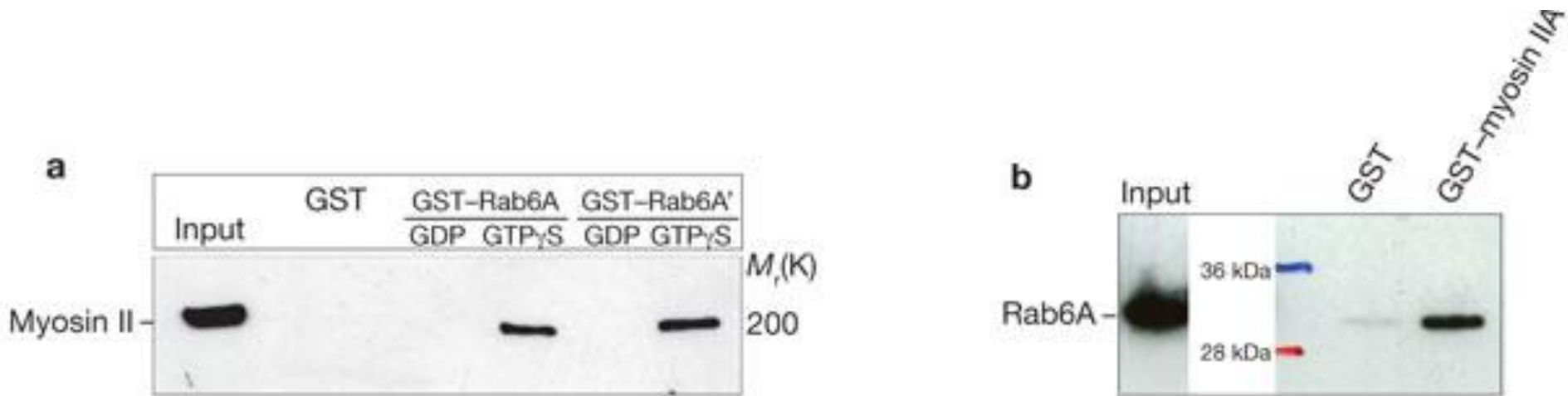
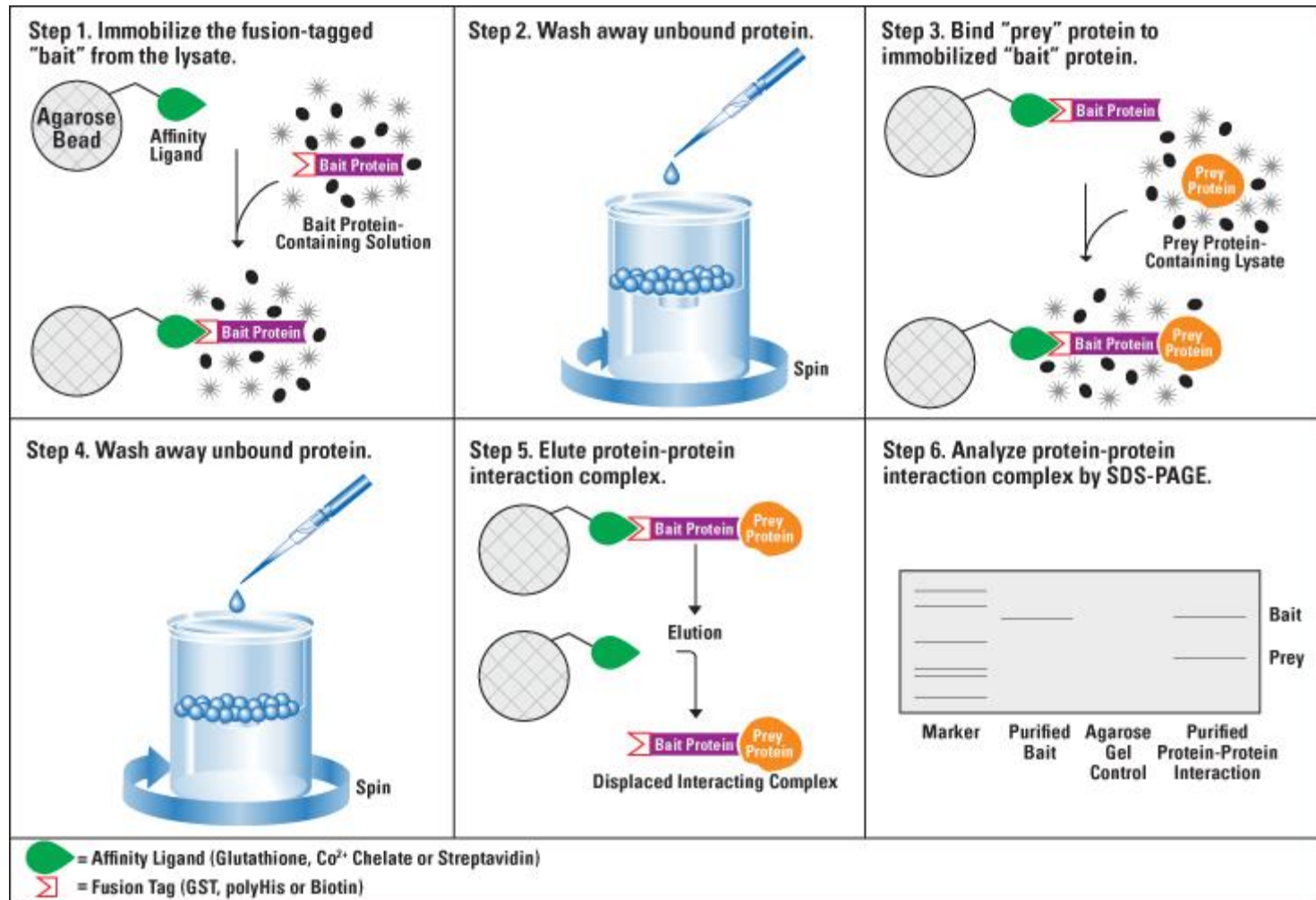
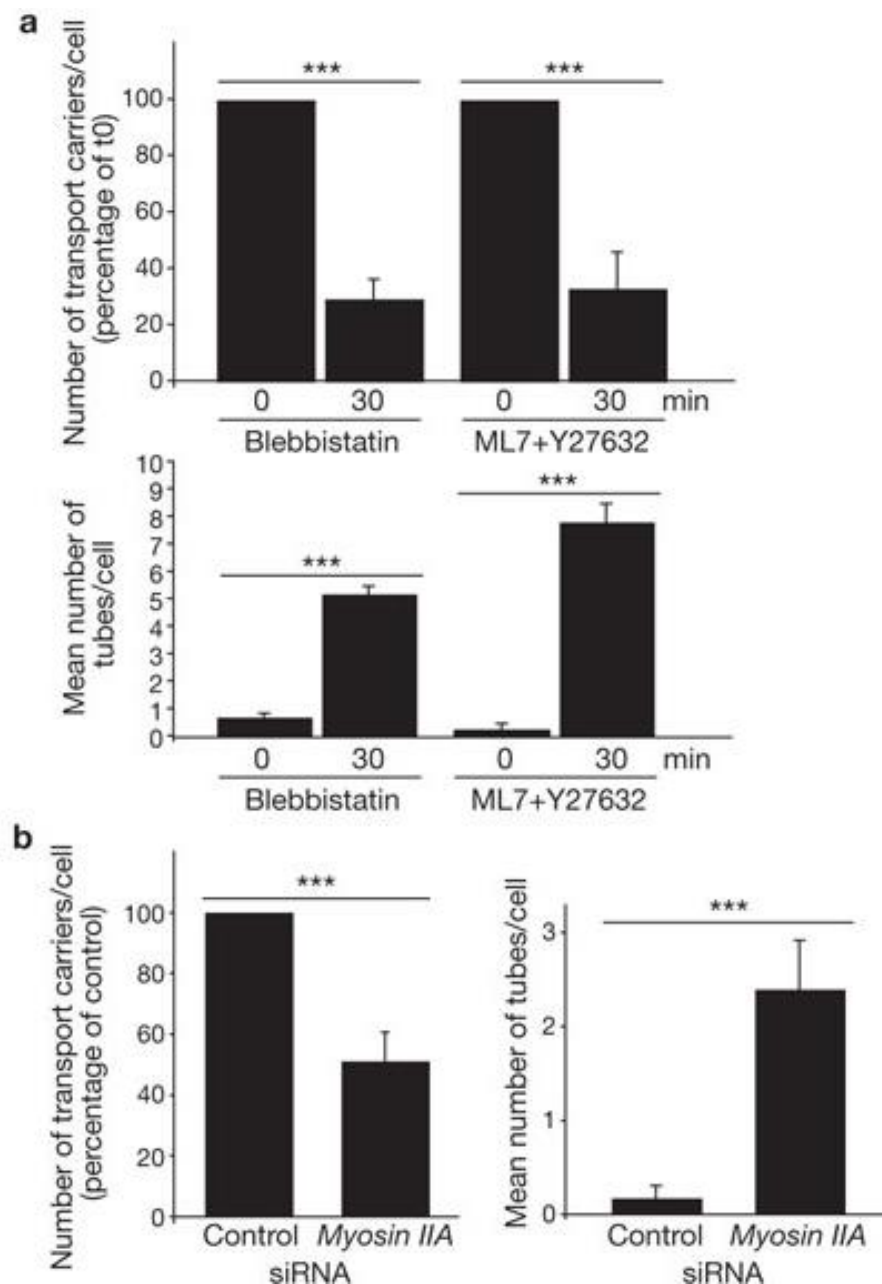


Figure 2. (a) Glutathione S-transferase (GST)–Rab6A or (GST)–Rab6A' fusion proteins were loaded with GDP or GTP γ S. After incubation with non-transfected NRK cells extracts, myosin II bound to either GST or GST–Rab beads was detected by western blot analysis using anti-myosin II antibody. Input represents a 5% load of the total cell extracts used in all conditions. (b) Purified GST–myosin IIA (amino acids 1147–1653) or GST (control) were incubated with purified His6–Rab6A loaded with GTP γ S. Input represents a 5% load of His6–Rab6A. Rab6A was detected by western blot analysis using an anti-Rab6 antibody.

Pull down assay



The authors then investigated the involvement of myosin II in the initiation step of Rab6-associated transport vesicle formation. For this, the dynamics of these vesicles was evaluated by fluorescence microscopy in HeLa cells. The results are shown in Figure 3.



(a) Quantification of the number of transport carriers and of Golgi-connected tubules in cells before (0 min) or 30 min after treatment with Blebbistatin (25 μ M) or (ML7 (30 μ M) + Y27632 (10 μ M)) to inhibit myosin II function. (mean $\pm$ s.e.m., n = 23–30 cells). (b) Quantification of the number of transport carriers and of Golgi-connected tubules after myosin IIA depletion (n = 24–37 cells). ***P < 10⁻⁵, Student's t-test.

Figure 3

Myosin II is a molecular motor that allows the contraction of actin microfilaments. To determine whether these microfilaments are involved in the formation of Rab6-associated vesicles, the authors performed the experiments presented in Figure 4 and 5.

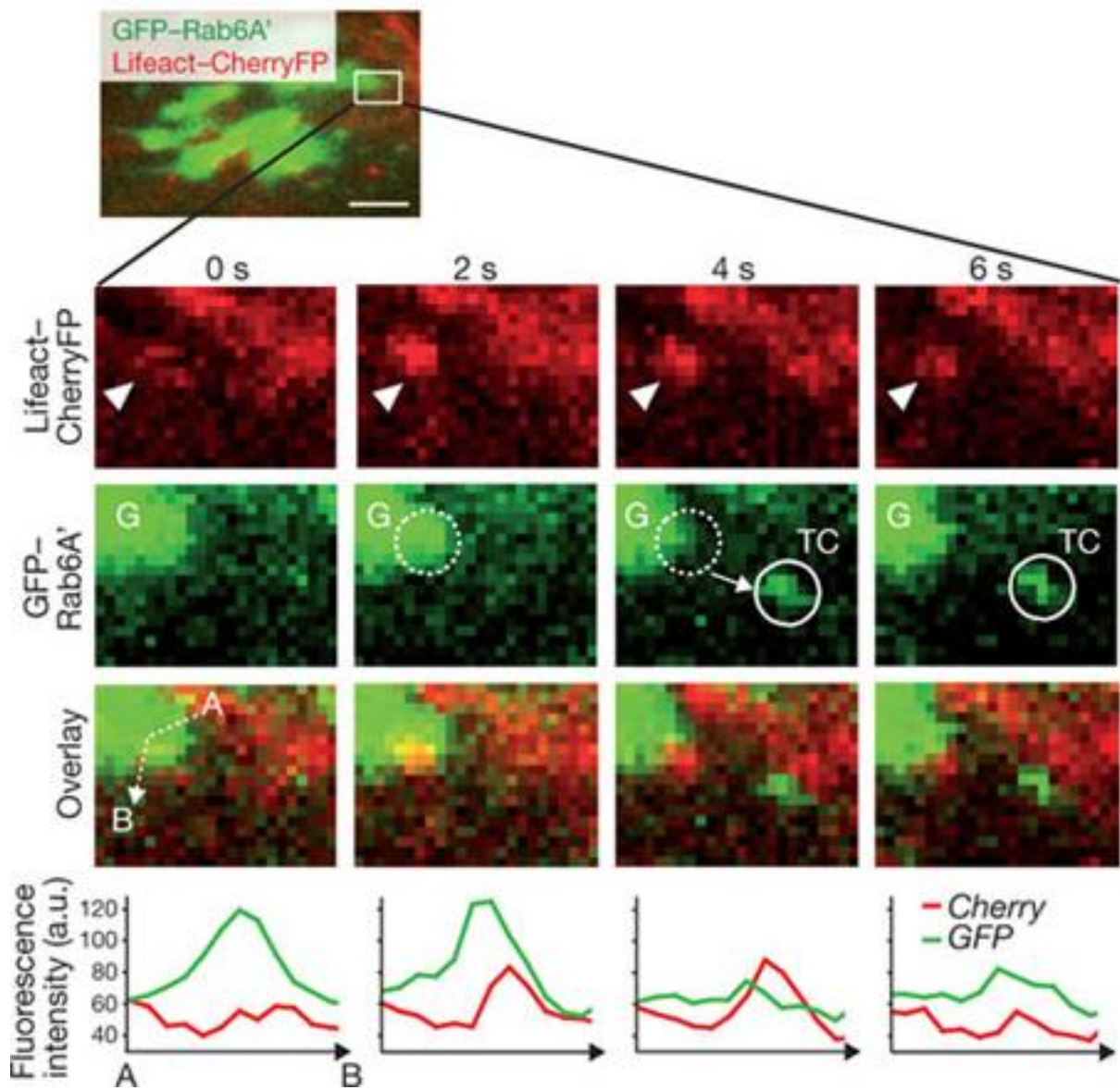
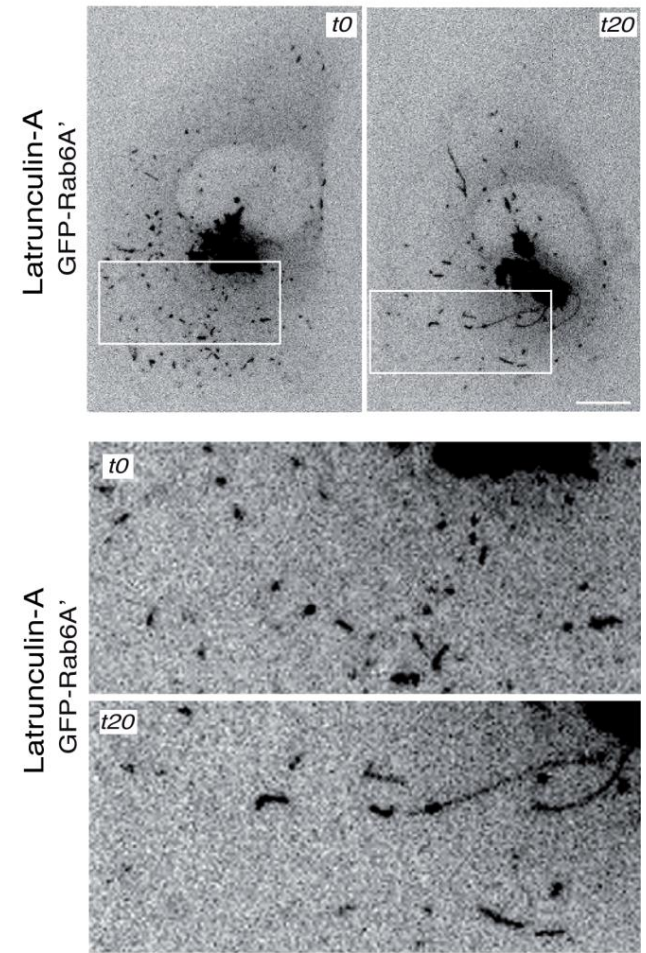


Figure 4. The upper panel displays a single confocal section of a HeLa cell co-expressing Rab6A-GFP (green) and the F-actin probe Lifeact-CherryFP (red). The middle panel shows high magnifications of a Golgi region (box) at the time of transport carrier fission. Snapshots of a 2 s interval time-lapse confocal movie are shown. Line profiles at each time point of the Rab6 (green) and the F-actin (red) fluorescence intensities (arbitrary units) in the a-b direction along the white dashed arrow centred to a fission event. Scale bar, 1 μ m.

Figure 4

Figure 5. (A) Representative frame of a HeLa cell expressing GFP-Rab6A' before and after a 20-min treatment with Latrunculin-A (0,5 μ M) to depolymerize F-actin. The middle panel shows high magnifications of a Golgi region (box). (B) Quantification of the number of Rab6-positive transport carriers and unfissioned Golgi-connected tubules before and after the treatment. Results are presented as mean $\pm$ SEM (n= 14 cells). ** $p < 10^{-3}$, *** $p < 10^{-5}$ (Student's t-test). Bar, 10 μ m.

A



B

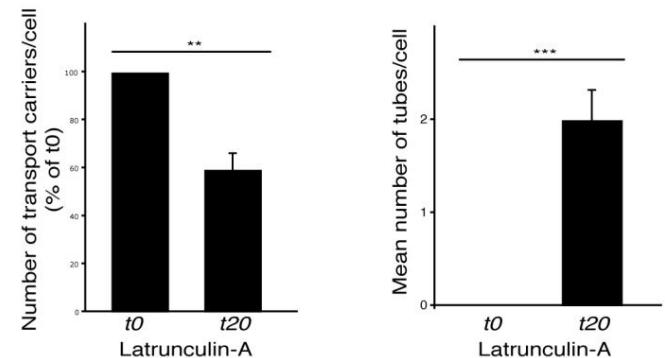


Figure 5

The Rab6-associated vesicles from the Golgi apparatus can be received in numerous cellular compartments, particularly at the level of the plasma membrane. In this context, the authors analyzed the transport of these vesicles to the plasma membrane using a marker specific for this trafficking (VSV-G). Their data are presented in Figure 6.

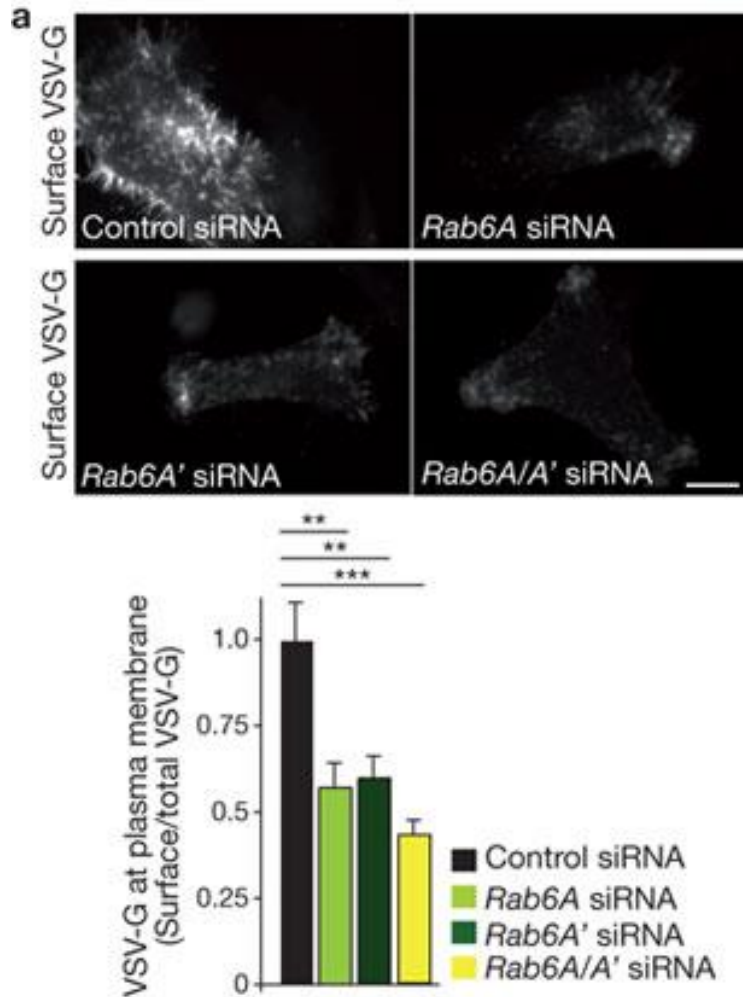


Figure 6. (a) HeLa cells expressing VSV-G–GFP tsO45 thermosensitive mutant were treated for 3 days with control, or siRNAs specific for Rab6A, Rab6A' or Rab6A/A', and were incubated overnight at 40°C then shifted to 32°C for 1.5 h to allow VSV-G trafficking. Cells stained for VSV-G at the plasma membrane by immuno-detection are shown. Quantifications of VSV-G at the plasma membrane (surface VSV-G to total VSV-G ratio) are indicated in the lower panel.

** $P < 5.10^{-2}$, *** $P < 5.10^{-3}$, Student's t-test (a–d). Scale bars, 10 μ m.

